Evolving Recurrent Neural Networks for Fault Detection and Prognostics

Abstract

This paper presents a novel data-driven fault prognostics methodology applied to rolling bearings and based on evolving granular models. Evolving granular models combines the flexibility of the incremental learning approaches usually employed for modelling data-streams with the potential for developing semantically relevant representations of granular computing. In particular, for dealing with fault prognostics, which involves nonlinear and time-varying degradation behavior, this paper proposes the evolving granular recurrent neural network (EGRNN) model. The EGRNN takes advantage of the simplified evolving fuzzy approach by incorporating the temporal memory capabilities of recurrent neural networks. In addition, the proposed approach employs the Principle of Justifiable Granularity for providing a granular uncertainty modelling. This integration enhances both the adaptability and explainability of the EGRNN, making it suitable fault prognostics.

Keywords: Fault Prognostics, Evolving Models, Granular Computing

1. Introduction

Predictive Maintenance (PdM) represents a strategic approach to monitoring industrial machinery throughout its life-cycle, focusing on enhancing reliability and safety while mitigating unscheduled downtime. Unlike traditional maintenance strategies, PdM utilizes data to identify incipient faults and anticipate failures, which minimizes the need for unnecessary manual inspections [1]. The core pillars of this methodology are Condition-Based Maintenance (CBM), concerned with the continuous acquisition and processing of data to identify the current state of a machine, and Prognostics and Health Management (PHM), a field aimed at detecting degradation, diagnosing faults, and predicting the Remaining Useful Life (RUL) [2].

Within the framework of CBM, fault detection and prognosis are often treated as distinct but sequential operations. Fault detection serves as the initial trigger, utilizing statistical methods or Machine Learning (ML) to identify deviations from normal operating behavior. Once a fault is detected, prognostics models – ranging from physics-based formulations to data-driven Artificial Intelligence (AI) [3, 4, 5] – are employed to model the degradation trajectory toward a defined failure threshold and estimate the RUL .

While AI-based approaches are favored for modeling complex nonlinear degradation patterns without explicit physical system knowledge, they face critical implementation barriers. Most conventional models require large datasets for offline training and remain static once deployed, failing to adapt to the non-stationary nature of degradation processes. This rigidity leads to performance decrease when encountering concept or context drift, often triggered by evolving operating conditions or novel system behaviors [6]. Moreover, model transparency remains a critical requirement for real-world prognostics. Traditional AI models often lack explainability, which hinders trust and limits their deployment in high-stakes maintenance decision-making processes [7, 8].

To address these limitations, this paper proposes the use of Evolving Granular Models (EGMs). While many computational intelligence models are intrinsically static and lack the capacity for continuous learning, Evolving Modelss (EMs) are characterized by their ability to autonomously expand or shrink their internal structure and incrementally tune parameters in response to non-stationary data streams. These systems transition into EGMs when integrated with the Granular Computing (GC) paradigm — a framework that processes data as collections of similar instances, known as Information Granules (IGs), which are grouped according to criteria such as proximity, similarity, or indistinguishability.

The synergy between evolving mechanisms and granular foundations makes EGMs suited for fault detection and prognostics. By inheriting the structural flexibility of EMs, these models can learn from scratch and adapt to non-stationary environments without requiring prior offline training. Simultaneously, the integration of GC allows the system to organize this knowledge into semantically meaningful IGs that represent specific operational patterns or health stages. In the context of machinery degradation, this dual capability enables the model to detect subtle changes in data patterns and associate active granules with different levels of equipment wear. Furthermore, the activation and creation of these granules provide a transparent trace of the

internal logic used to generate a prediction, granting the model a high degree of interpretability and explainability — critical factors for trust in machinery failure detection.

Despite their potential, the application of EGMs to simultaneous fault detection and RUL forecasting remains relatively underexplored. Works such as [9, 10, 11, 12] achieved accurate RUL predictions with different granular approaches, but none of them concerned with multiple tasks execution.

Contributions. This paper presents the Evolving Granular Recurrent Neural Network (EGRNN), a model combining Tipicality and Eccentricity Data Analytics (TEDA) and the Recurrent Neural Network (RNN) framework applied to execute simultaneously fault detection and prognostic RUL of rolling bearings. This model takes advantage of the simplified fuzzy approach provided by TEDA to cluster data into distinct IGs associated with health stages and failure degradation patterns and also ensure predictions under an interpretable framework and the temporal memory capabilities of RNNs to retain and exploit historical information embedded in time series data, enabling it to track gradual patterns of variations and capture evolving system dynamics.

- An EGM founded on TEDA-based algorithms, capable of dynamically expanding or shrinking its structure and adjusting parameters based on the knowledge learned from the data, while clustering data into IGs representative of health stages and failure modes.

- The integration of RNNs with each IG to address both nonlinearity and temporal dynamics more effectively than traditional linear modeling methods. By handling nonlinearity and temporal dynamics more effectively, the RNN-based model presents a potentially better option for tackling complex real-world problems with time-dependent data.

- The proposed EGRNN is integrated with granular uncertainty quantification to establish an original RUL prediction framework for failure prognostics, efficacious in managing the uncertain and nonlinear behaviors inherent in degradation processes.

2. Preliminaries

2.1. Notations

This work used some notations whose descriptions are listed at Table 1.

Table 1: Notations list.

Notation	Description
h	Iteration counter
i	Quantity counter
$\mathcal{G}_i$	Information granule n° i
$\mathcal{P}_{i,h}$	Granule n° i parameters at iteration h
X_h	Regression data input at instant h
Z_h	Granulation data input at instant h
Y_h	Real data output at instant h
$\hat{Y}_h$	Predicted singular data output at instant h
$\hat{Y}_{i,h}$	Predicted granular data output at instant h

2.2. Evolving Granular Models

Consider an initially empty EGM; Figure 1 illustrates the data flow and how granulation as inputs Z and regression inputs X are associated to information granules IGs and transformed into outputs $\hat{Y}$ through the model framework across three distinct time subsequent non consecutive instants h, i, and j.

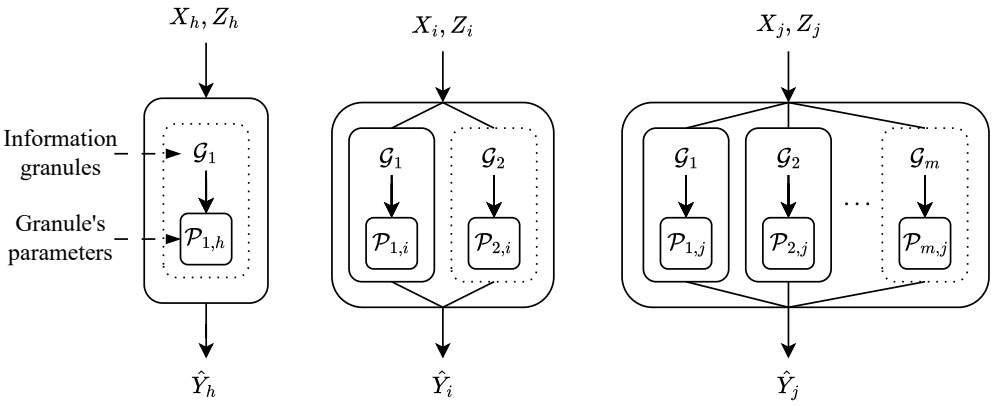


Figure 1: Data flow through an EGM framework.

Because at first iteration $h = 1$ the model operates without prior information, the assimilation of the initial input X_h activates the learning mechanism, triggering the autonomous creation of the first information granule $\mathcal{G}_1$.

Simultaneously, a set of knowledge parameters $\mathcal{P}_{1,h}$ is assigned to $\mathcal{G}_1$ to characterize the gathered information at that instant. Following this structural adaptation, the model calculates and provides its first numeric output Y_h.

The EGM accommodates subsequent data inputs within the existing structure. Until iteration $i-1$, incoming samples X and their information are assimilated by $\mathcal{G}_1$, which involves the iterative adjustment of its associated knowledge parameters $\mathcal{P}_1$ to reflect the evolving data distribution. However, at iteration i, the model encounters a data sample X_i sufficiently distinct from the information gathered by $\mathcal{G}_1$, preventing its accommodation. This triggers the creation of the second granule $\mathcal{G}_2$, with its respective set of parameters $\mathcal{P}_{2,i}$. Consequently, the model's predictive capability is expanded, allowing it to calculate Y_i by considering the contribution of the new IG.

From iteration $i+1$ onward, the model continues to assimilate consecutive data inputs X, expanding its knowledge base by dynamically adapting both its parameters and its structure in response to the incoming data stream. This ongoing evolution allows the EGM to maintain an optimized representation of the system's current state, ensuring that each new sample contributes to the refinement of existing granules or the emergence of new ones as novel patterns appear.

Throughout this recursive process, the model calculates outputs based on the collective information held within the current set of IGs $\{\mathcal{G}\}_h$ and their respective set of parameters $\{\mathcal{P}\}_h$. The predicted output $\hat{Y}_h$ of an EGMs, established at (5), can be represented by an unknown time-dependent function f_h of the inputs X_h, Z_h, $\{\mathcal{G}\}_h$ and $\{\mathcal{P}\}_h$ at instant h.

$$\hat{Y}_h = \mathrm{f}_h\left(X_h, Z_h, \{\mathcal{G}\}_h, \{\mathcal{P}\}_h\right) \tag{1}$$

By simultaneously and recursively updating its internal architecture ($\{\mathcal{G}\}_h$ and $\{\mathcal{P}\}_h$) and generating real-time predictions, an EGM effectively tracks the dynamic variations and non-stationary behaviors inherent in degradation processes. During each iteration the parameters are tuned and $\{\mathcal{G}\}_h$ may change as a function $f_{\mathcal{G}}$ and $f_{\mathcal{P}}$ of themselves at the previous iteration $h-1$ and the current inputs X_h and Z_h.

$$\{\mathcal{G}\}_h = \mathrm{f}_{\mathcal{G}}\left(X_h, Z_h, \{\mathcal{G}\}_{h-1}\right) \tag{2}$$

$$\{\mathcal{P}\}_h = \mathrm{f}_{\mathcal{P}}\left(X_h, Z_h, \{\mathcal{P}\}_{h-1}\right) \tag{3}$$

Consider two data inputs, Z_h for granulation and X_h for forecasting, a generic regression function f and its output Y_h. An EGM follows if-then

rules established as (4).

$$\text{Rule } i: \text{ If } Z_h \text{ is } \mathcal{G}_i, \text{ then } \hat{Y}_{i,h} = f_{i,h}\left(X_h\right) \tag{4}$$

where, given inputs X_h and Z_h each regression function $f_{i,h}$, associated to its respective IG $\mathcal{G}_i$, at an instant h, calculates a granular output $\hat{Y}_{i,h}$. Then, the singular output $\hat{Y}_h$ is calculated through a weighted sum as (5)

$$\hat{Y}_h = \sum_{i=1}^{n} w_{i,h} \hat{Y}_{i,h} \tag{5}$$

where, $w_{i,h}$ represents the weight of input Z_h at granule $\mathcal{G}_i$.

2.2.1. Principle of Justifiable Granularity

In the context of fault detection, IGs serve as semantically meaningful structures that represent distinct health stages or operational patterns of a system. By grouping similar data instances, IGs support human interpretability by providing a transparent view of how the model identifies degradation. However, a major challenge in fault prognostics is ensuring that each granule captures only meaningful information; excessive or irrelevant data can compromise both the accuracy of health stage identification and the model's interpretability.

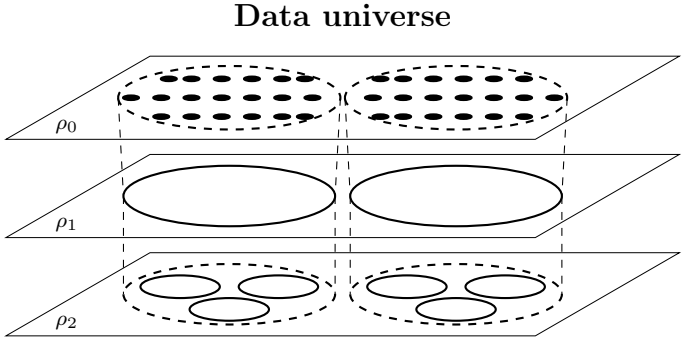


Figure 2: Information granule structure and granularity.

To address this, the significance of a granule $\mathcal{G}_i$ is evaluated through the Principle of Justifiable Granularity (PJG), which determines the level of

information abstraction within the domain space. Figure 2 illustrates this concept, where data samples can be organized into various granularities ρ to capture different numbers of granules with varying sizes. To support a coherent creation of IGs, [13] proposed the PJG based on:

1. **Experimental evidence:** A granule $\mathcal{G}_i$ must have sufficient coverage $\overline{\text{cov}}_i$, meaning it is supported by a significant cardinality of samples X.

$$\overline{\text{cov}}_i = \frac{1}{N}\sum_h C_h \quad \text{where} \quad C_h = \begin{cases} 0, & X_h \notin \mathcal{G}_i \\ 1, & X_h \in \mathcal{G}_i \end{cases} \tag{6}$$

2. **Semantic meaning:** A granule $\mathcal{G}_i$ must possess high specificity $\overline{sp}_i$, meaning that the size S occupied by its information is compact enough to distinguish between different IGs.

$$\overline{\text{sp}}_i = 1 - \frac{S\left(\mathcal{G}_i\right)}{S_{\max}\left(\mathcal{G}_i\right)} \tag{7}$$

A justifiable granule must concentrate the largest amount of data possible—coverage— inside the smallest size possible—specificity. This leads us to a two-objective optimization problem (maximization) with respect to ρ, or we can transform it into a scalar optimization problem by maximizing the product of coverage and specificity [14], viewing it as a function of ρ as follows.

$$\mathcal{G}\left(\rho\right) = \overline{\text{cov}} \cdot \overline{\text{sp}} \quad \text{and} \quad \rho_{\text{opt}} = \arg\max_{\rho} \mathcal{G}\left(\rho\right) \tag{8}$$

2.3. Problem Formulation

The challenge in developing an EGM capable of both fault detection and prognostics lies in modeling a learning mechanism capable of accurately estimating RUL and creating a quantity of IGs that conciliate the health stages (specificity) and their duration (coverage).

Fault Detection:

Let the values that specify the health stage of an equipment be called Degradation Stage Indicators (DSIs) and the time series that tracks the equipment health stage over time $\text{DSI} = z_1, \ldots, z_h, \ldots, z_H$.

Considering that during the machinery lifetime there must be at least two health stages to differentiate its healthy and faulty stages, the first main

objective is to obtain a clustering algorithm capable of autonomously partitioning the DSI into n IGs as follows.

$$\mathcal{G}_i \subset \text{DSI} \,|i = 1, \ldots, n \text{ where } n \geq 2 \tag{9}$$

$$z_h = \mathcal{G}_i \,|\text{P}_i\,(z_h) \leq \text{T}\,(\mathcal{G}_i) \tag{10}$$

where P_i denotes a compatibility function quantifying the relevance of z_h to $\mathcal{G}_i$, and T represents the dynamic threshold associated with the granule's boundaries.

Under this formulation, fault detection is achieved when a sample z_h fails to satisfy the inclusion criteria for the "healthy" granule $\mathcal{G}_1$, thereby triggering the creation of a subsequent granule $\mathcal{G}_2$ associated with an incipient fault stage.

RUL Prediction: Let the values that specify the healthiness of an equipment be called Health Indicators (HIs) and the time series that tracks the equipment health over time $\text{HI} = x_1, \ldots, x_h, \ldots, x_H$. The second main objective is to obtain a multi-step-ahead predictor of the form:

$$\hat{x}_{h+H} = \text{f}_\text{g}\left(\{\hat{x}_{h_i}\}_{h_i \leq h}\right) \tag{11}$$

which, a generic function or model f_g provides the next predictions of the HI H steps ahead, given the HI series up to the current instant h.

The RUL estimation is based on predictions of the HI time series. Assuming that the health indicator HI is sufficiently monotonic and increasing (i.e., x_h increases as the end of life approaches), there exists a threshold related to the End of Life (EoL) of the equipment, i.e., for $x_h \geq \text{EoL}$, the equipment's useful life has ended. Considering that a given piece of equipment reaches the end of its life at time H, the real RUL_h and predicted $\hat{\text{RUL}}_h$ remaining life at time $h \leq H$ are respectively defined in (12).

$$\text{RUL}_h = H - h, \quad \hat{\text{RUL}}_h = \inf\left\{\hat{H} \in \mathbb{N} : \hat{x}_{h+\hat{H}} \geq \text{EoL}\right\} \tag{12}$$

3. Evolving Recurrent Neural Models

3.1. Typicality and Eccentricity Data Analytics

Built upon the concepts of typicality and eccentricity, TEDA is a recursive, rule-based learning framework that evaluates data instances based on their similarity and dissimilarity. A significant advantage of this approach

is its independence from prior data knowledge or the need for predefined parameters.

Given a time series of consecutive data instances $\mathrm{X} = \{X_h \mid h \in \mathbb{N}\}$, the TEDA framework characterizes each sample through two complementary metrics: eccentricity and typicality. The eccentricity ξ_h, which is bounded within the interval $[0, 1]$, quantifies the degree of dissimilarity between the current sample X_h and the cumulative historical database. Conversely, the typicality τ_h reflects the similarity of X_h to the existing data. These values are formally defined as follows:

$$\xi_h = \frac{\sum_{j=1}^{h} d(X_h, X_j)}{\sum_{j=1}^{h} \sum_{k=j+1}^{h} d(X_j, X_k)} \tag{13}$$

$$\tau_h = 1 - \xi_h \tag{14}$$

The operator $d(\cdot)$ denotes the distance between elements X_j and X_k. When employing the Euclidean distance, ξ_h can be computed through the following recursive formulations:

$$\xi_h = \frac{1}{s_h} + \frac{(\mu_h - X_h)^\top (\mu_h - X_h)}{s_h \sigma_h^2} \tag{15}$$

$$s_h = s_{h-1} + 1 \tag{16}$$

$$\mu_h = \mu_{h-1} \frac{(s_h - 1)}{s_h} + \frac{X_h}{s_h}, \tag{17}$$

$$u_h = u_{h-1} \frac{(s_h - 1)}{s_h} + \frac{X_h^\top X_h}{s_h}, \tag{18}$$

$$\sigma_h^2 = u_h - \mu_h^\top \mu_h. \tag{19}$$

Where s_h represents the cumulative cardinality of the database, while μ_h and σ_h^2 denote the recursive mean and variance up to time h, respectively. Additionally, u_h is defined as the mean inner product of the input samples. Collectively, these variables constitute the set of IG's parameters $\mathcal{P}_h = \{\xi_h, \tau_h, s_h, u_h, \sigma_h^2\}$.

Within the TEDA framework, data is organized into granules $\mathcal{G}_i$ whose spatial coverage of each granule is defined by the boundary in Inequality (20), where m represents a granular sensitivity factor—a parameter that regulates the model's structural evolution by controlling the density and total number

of granules generated from the data stream.

$$\xi_h \leq \frac{m^2+1}{s_h} \tag{20}$$

In a TEDA-based framework comprising multiple granules $\{\mathcal{G}_i, \mathcal{G}_j, \ldots\}$, the recursive parameters, (13) to (20), become specific to each individual IG. Consequently, these variables are indexed by the granule's identity i; for instance, $\xi_{i,h}$ denotes the local eccentricity of the i-th granule $\mathcal{G}_i$ at the discrete time instant h.

3.2. AutoCloud

Founded on TEDA, [15] is an evolving granular clustering algorithm that allows a single data sample X_h to be simultaneously covered by multiple IGs. To manage this overlap and ensure a compact model structure, the model incorporates a mechanism for merging granules that share a significant portion of data. This fusion process is triggered when the intersection of two granules satisfies the conditions defined in (21).

$$s_h^{\mathcal{G}_i \cap \mathcal{G}_j} > s_{ih} - s_h^{\mathcal{G}_i \cap \mathcal{G}_j} \text{ and/or } s_h^{\mathcal{G}_i \cap \mathcal{G}_j} > s_{jh} - s_h^{\mathcal{G}_i \cap \mathcal{G}_j} \tag{21}$$

When the conditions (21) are met, the parameters of the resulting information granule $\mathcal{G}_r$ are calculated through (22)–(25). This mechanism ensures that the model can autonomously adapt to the data stream by not only expanding its structure through the creation of new granules but also contracting it through the merger of redundant ones.

$$s_{rh} = s_{ih} + s_{jh} - s_h^{\mathcal{G}_i \cap \mathcal{G}_j} \tag{22}$$

$$\mu_{rh} = \frac{s_{ih}\mu_{ih} + s_{jh}\mu_{jh}}{s_{ih} + s_{jh}} \tag{23}$$

$$u_{rh} = \frac{s_{ih}u_{ih} + s_{jh}u_{j_h}}{s_{ih} + s_{jh}} \tag{24}$$

$$\sigma_{rh}^2 = \frac{(s_{ih}-1)\sigma_{ih}^2 + (s_{jh}-1)\sigma_{jh}^2}{s_{ih} + s_{jh} - 2} \tag{25}$$

3.3. Recursive Learning Recurrent Neural Networks

RNNs are widely utilized for processing sequential data and time series. In contrast to traditional feedforward Neural Networkss (NNs), RNNs incorporate a feedback loop that provides a memory mechanism, enabling them

to retain information from past inputs—a feature essential for tasks where temporal order is critical.

Typically, its weight matrices (U, V, and W) are adjusted during a static, offline training phase using gradients derived via the Back-propagation Through Time (BPTT) technique. However, integrating an RNN into the framework of an EM requires the adaptation of its learning mechanism to support recursive learning.

To handle this, the Random Feedback Local Online (RFLO) was developed as a learning mechanism for RNNs [16]. The technique is a gradient-based approach that was adapted to depend only on local information about the input and output, in addition to a random feedback projection of the RNN output error. The following equations define the proposed Recursive Learning Recurrent Neural Network (RLRNN) in terms of its inputs X_h, predicted outputs $\hat{Y}_h$. Where a_h is the total input, h_h is the activity vector, and e_h is the prediction error.

$$a_h = \tanh\left(UX_h + V\mathrm{h}_{h-1}\right) \tag{26}$$

$$\mathrm{h}_h = \mathrm{h}_{h-1} + \frac{1}{\mathrm{t}}[-\mathrm{h}_{h-1} + \tanh\left(a_h\right)] \tag{27}$$

$$\hat{Y}_h = W\mathrm{h}_h \tag{28}$$

$$e_h = Y_h - \hat{Y}_h \tag{29}$$

The gradient-based learning mechanism RFLO is defined by the following weight-update equations. Where η are the learning rates, B is a random matrix of feedback weights, p_h and q_h are the accumulated products of the pre- and postsynaptic activity at the recurrent and input synapses, respectively.

$$\Delta W_h = \frac{1}{T}\sum_{t=1}^{h} \eta_1 e_t \mathrm{h}_t \tag{30}$$

$$\Delta V_h = \frac{1}{T}\sum_{t=1}^{h} \eta_2 [\mathrm{B}\cdot e_t] p_t \tag{31}$$

$$\Delta U_h = \frac{1}{T}\sum_{t=1}^{h} \eta_3 [\mathrm{B}\cdot e_t] q_t \tag{32}$$

$$p_h = \frac{1}{\mathrm{t}}\tanh\left(a_{h-1}\right)\mathrm{h}_{h-1} + \left(1-\frac{1}{\mathrm{t}}\right)p_{h-1} \tag{33}$$

$$q_h = \frac{1}{\mathrm{t}}\tanh\left(a_{h-1}\right)X_{h-1} + \left(1-\frac{1}{\mathrm{t}}\right)q_{h-1} \tag{34}$$

3.4. Evolving Granular Recurrent Neural Networks

The proposed EGRNN extend clustering ability of AutoCloud by improving the semantic meaning of each IG applying PJG approach and also adding prediction capabilities through the previously presented RLRNN.

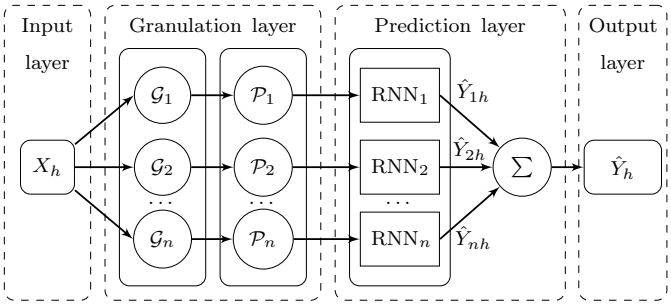


Figure 3: Evolving recurrent neural network framework.

Figure 3 depicts the EGRNN framework, where a given data sample X_h goes through the granulation layer, triggering the granule manipulation policies, crosses the granular prediction layer where each IGs's EGRNN estimate a granular output $\hat{Y}_{ih}$ and aggregates them by a weighted sum denoted by the $\sum$ operator, and finally provides the singular output $\hat{Y}_h$.

The typicality τ associated to TEDA (AutoCloud's core) is analogous to to fuzzy membership degree in fuzzy sets, allowing the composition of the

rule-based model ahead, where $i \in \mathbb{N}^*$, $X_h = (x_{h-l}, \ldots, x_h)$ is a data window containing the current sample and l past samples and RNN_{ih} is the EGRNN associated to each granule $\mathcal{G}_i$.

$$\text{Rule } i: \text{ If } X_h \text{ is } \mathcal{G}_i, \text{ then } \hat{Y}_{ih} = \text{RNN}_{ih}\left(X_h\right) \tag{35}$$

The singular output prediction $\hat{Y}_h$ is the sum of each $\mathcal{G}_i$'s prediction $\hat{Y}_{ih}$ pondered by its weight $w_{ih} = \frac{\tau_{ih}}{\sum_{j=1}^{n} \tau_{jh}}$

$$\hat{Y}_h = \sum_{i=1}^{n} w_{ih} \hat{Y}_{ih} \tag{36}$$

Because AutoCloud has no prediction step, it is proposed that when two granules $\mathcal{G}_i$ and $\mathcal{G}_j$ are merged, the values of the resulting RNN_{rh} weights matrix U_{rh}, V_{rh}, and W_{rh} are computed by a weighted average based on each granule's typicality.

$$W_{rh} = \frac{\tau_{ih} W_{ih} + \tau_{jh} W_{jh}}{\tau_{ih} + \tau_{jh}},\ V_{rh} = \frac{\tau_{ih} V_{ih} + \tau_{jh} V_{jh}}{\tau_{ih} + \tau_{jh}},\ U_{rh} = \frac{\tau_{ih} U_{ih} + \tau_{jh} U_{jh}}{\tau_{ih} + \tau_{jh}} \tag{37}$$

3.5. Justifiable Granules Modeling

Because the eccentricity established at (15) considered the Euclidian approach to measure the distance between samples, the space occupied by a IG, established at (7), assumes the form of a n-dimensional volume of a Euclidean sphere of radius R as follows.

$$V_n(R) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)} R^n \tag{38}$$

where Γ is Euler's gamma function and n is the dimension of the input sample.

3.6. Granular Uncertainty Modeling

The concept of granularity can be applied to granular models to obtain granular predictions by elevating their output to a granular counterpart with a granularity of ρ, which also yields granular predictions.

The approach proposed by [14] makes the output granular by building an interval around the result produced by the numeric model. The procedure approximates the prediction error e_h as a function E —polynomial by [14]—

of the model's output $\hat{Y}_h$ and estimates the prediction errors $\hat{e}_h = \mathrm{E}(\hat{Y}_h)$. Then the product $\rho \cdot \hat{e}_h$ is used as the output's margin error, which leads $\hat{Y}_h$ to the minimum and maximum interval $[\underline{Y}_h, \overline{Y}_h]$ as follows.

$$[\underline{Y}_h, \overline{Y}_h] = [\hat{Y}_h - \rho \cdot \hat{e}_h, \hat{Y}_h + \rho \cdot \hat{e}_h] \tag{39}$$

Instead of a polynomial function, this work proposes a logarithmic function to approximate the prediction error, adapting the methodology described above. This function can be modeled using the Recursive Least Squares with Variable-Direction Forgetting (RLS-VDF) proposed by [17].

4. Fault Prognostics based on EGRNN

4.1. An Overview of Fault Prognostics and RUL Prediction

Fault prognostics is carried out through the consecutive execution of the steps shown in the diagram in Figure 4, which will be explained in detail throughout this section. As illustrated in Figure 4, the main steps of a prognostics algorithm are: data acquisition and pre-processing; feature extraction and selection; computation of a health index; and RUL prediction. Those steps are discussed below.

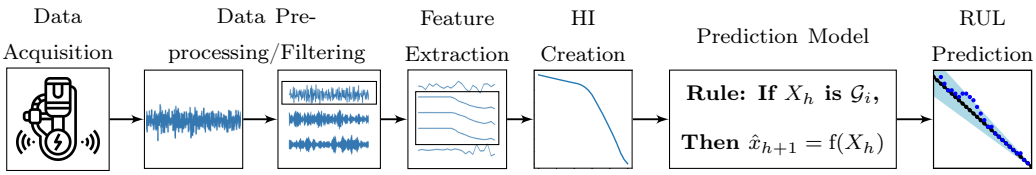


Figure 4: Key steps of fault prognostics.

Data Acquisition and Pre-processing. Data acquisition involves recording and storing measurements of operational parameters from the analyzed equipment. However, during this stage, noise and other disturbances are also introduced into the data, diluting its relevant information. Subsequently, data preprocessing is performed to filter the data and highlight only the pertinent information. Methods such as moving average and exponential smoothing can be used, as well as more advanced signal processing techniques like Variational Mode Decomposition (VMD) and wavelet transform.

Feature Extraction and Selection. After processing, feature extraction is performed, typically involving statistical indicators calculated from

data samples in both the time and frequency domains, which can describe the degradation of the equipment. The best features are selected based on indicators that quantify their relevance. Since real machines exhibit progressive and irreversible wear, increasing features over time are preferred. These are evaluated using indicators such as monotonicity and trendability [18].

Consider a time series $\mathcal{C} = \{\mathcal{C}_h \in \mathbb{R}^1 \mid h \in \mathbb{N}^*\}$ as a candidate health index, indexed by h relative to the time instances belonging to the sampling sequence $\mathbb{H} = h_i; , i \in \mathbb{N} \leq K$. The monotonicity $\mathrm{M}(\mathcal{C})$ and trendability $\mathrm{T}(\mathcal{C})$ of this series are defined as:

$$\mathrm{M}(\mathcal{C}) = \frac{1}{K-1} \left| \sum_{h \in \mathcal{H} \setminus \{h_N\}} \operatorname{sgn}\left(\mathcal{C}_h - \mathcal{C}_{h+1}\right) \right|, \tag{40}$$

$$\mathrm{T}(\mathcal{C}, \mathcal{H}) = \operatorname{corr}\{\mathcal{C}, \mathcal{H}\}. \tag{41}$$

where sgn is the sign function, corr is the Pearson correlation coefficient, and K is the length of the time series $\mathcal{C}$. Monotonicity measures how consistently increasing or decreasing a series is, while trendability measures the correlation between the feature and time.

Construction of the Health Indicator. HIs are defined as numerical values capable of quantifying the level of degradation of a piece of equipment. HIs can be physical when they have a mathematical, physical, or statistical foundation, or virtual when they are formed by the fusion of multiple physical HIs [19]. More generally, for a set of N features $\bar{\mathcal{C}}_1, \ldots, \bar{\mathcal{C}}_N$, a health indicator HI is created by combining various features, that is, $\mathrm{HI} = T(\bar{\mathcal{C}}_1, \ldots, \bar{\mathcal{C}}_N)$. Among the methods for feature fusion, Principal Component Analysis (PCA) stands out.

Construction of the Degradation Stage Indicator. DSIs are defined as numerical values capable of associate the HI to a stage with similar degradation trend [19]. Consider V_h an operational parameter time series signal $\mathrm{V}_g = \{v_d \in \mathbb{R}^1 \mid d \in \mathbb{N}^*\}$ sampled at a time instant g with d points and $\mathcal{V} = \{\mathrm{V}_h \mid h \in \mathbb{N}^*\}$ the set of consecutively sampled signals V, the DSI time series $\mathrm{Z} = (z_1, \ldots, z_N)$ can be calculated by the Pearson Correlation Coefficient (PCC) denoted by z as follows.

$$z_g = \frac{\sum_{a=1}^{b} \left(\mathrm{V}_g - \overline{\mathrm{V}}_g\right)\left(\mathrm{V}_1 - \overline{\mathrm{V}}_1\right)}{\sqrt{\sum_{a=1}^{b} \left(\mathrm{V}_g - \overline{\mathrm{V}}_g\right)^2 \left(\mathrm{V}_1 - \overline{\mathrm{V}}_1\right)^2}} \tag{42}$$

RUL Prediction. After designing the HI, the next step in fault prognosis is the construction of a multi-step-ahead predictor. It should be noted that many prognostic studies use predictors of HI time series to indirectly estimate RUL. For a time series HI $= x_1, \ldots, x_N$, the goal is to obtain a predictor of the form:

$$\hat{x}_{h+N} = \mathrm{f}_r\left(\{\hat{x}_{h_i}\}_{h_i \leq h}\right), \tag{43}$$

which, a generic function / model f_r, at instant h, provides the next prediction of the HI N steps ahead, given the HI series up to the current time h.

A key task of prognostics is the RUL estimation based predictions of the HI time series. Assuming that the health index HI is sufficiently monotonic and increasing (i.e., x_h increases as the end of life approaches), there exists a threshold related to the EoL of the equipment, i.e., for $x_h \geq$ EoL, the equipment's useful life has ended. Considering that a given piece of equipment reaches the end of its life at time H, the RUL at time $h \leq H$ is defined as:

$$\mathrm{RUL}_h = H - h. \tag{44}$$

On the other hand, the RUL prediction obtained through the predictor (43) is defined as:

$$\hat{\mathrm{RUL}}_h = \inf\{N \in \mathbb{N} : \hat{x}_{h+N} \geq \mathrm{EoL}\}. \tag{45}$$

4.2. EGRNN for RUL Prediction

The EGRNN usage in RUL prediction implies some modifications on equations (35)-(36). As machinery degradation presents some different patterns during its life cycle, it's possible to establish a relationship between the HI time series and a Degradation Stage (DS) time series, which can be interpreted as a Premise Variable (PV)—and create DS-based IGs while using HI to predict, instead of both granulating and predicting with the same series.

Considering a HI time series $\mathrm{X} = \{x_t \in \mathbb{R}^{\mathrm{n}} \mid t \in \mathbb{N}^*\}$, a HI data sample $X_h = (x_{h-l}, \ldots, x_h) \mid X_h \in \mathrm{X}$, a DS time series $\mathrm{Z} = \{z_t \in \mathbb{R}^{\mathrm{n}} | \ t \in \mathbb{N}^*\}$, and a DS data sample $Z_h = (z_{h-l}, \ldots, z_h) \mid Z_h \in \mathrm{Z}$, the predictor must calculate the output $\hat{Y}_h = \hat{x}_{h+1}$ through X_h. So, the fuzzy-based rule established by (35) is modified as follows.

$$\text{Rule } i: \text{ If } Z_h \text{ is } \mathcal{G}_i, \text{ then } \hat{Y}_{ih} = \hat{x}_{h+1} = \mathrm{RNN}_{ih}\left(X_h\right) \tag{46}$$

By creating granules $\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_n$ associated with specific DSs, a given DS Z_h generates different typicalities $\tau_{1h}, \tau_{2h}, \ldots, \tau_{nh}$, with higher values for

the granules $\mathcal{G}$ that are most closely related to the current DS. This process leads to an increase in typicality—indicating the relevance of Z_h—for the granules associated with the current DS, while simultaneously decreasing the typicality for granules associated with other DSs.

Considering that granular predictions $\hat{Y}_{ih} = \mathrm{RNN}_{ih}(X_h)$ are pondered by a typicality weight w_{ih}, each product $w_{ih} \cdot \hat{Y}_{ih}$ can be interpreted as a relevance-weighted contribution of each singular prediction to the overall output. The magnitude of each contribution depends directly on its corresponding typicality weight w_{ih}. This approach ensures that the singular output $\hat{Y}_h$ predominantly reflects the most relevant singular predictions.

4.3. Uncertainty quantification for Prognostics

Consider a generic variable $\mathcal{V}$ bounded by a minimum and maximum value $\underline{\mathcal{V}} \leq \mathcal{V} \leq \overline{\mathcal{V}}$ the notation $[\mathcal{V}]$ denotes the interval $[\mathcal{V}] = [\underline{\mathcal{V}}, \overline{\mathcal{V}}]$. By applying the procedure outlined in Section 3.6 and given by (39), a singular output $\hat{Y}$ of the EGRNN is transformed into an interval output $[\hat{Y}] = [\underline{Y}, \overline{Y}]$.

This transformation allows the RUL prediction to be represented as a range, providing a prediction within an interval rather than a single value. Approximating the HI's prediction error as a logarithmic function of the model's output enables the calculation of a margin error $\mathrm{M_h}$ around $\hat{Y}_h$. Consider a current input $X_h = (x_{h-l}, \ldots, x_h)$ and its respective output $\hat{Y}_h = x_{h+1}$, the HI prediction's margin error $\mathrm{M_h}$ is calculated as follows.

$$\hat{e}_h = \theta_1 + \theta_2 \cdot \log(\hat{Y}_h) \tag{47}$$

$$\mathrm{M_h} = \rho \cdot \hat{e}_h \tag{48}$$

Where θ_1 and θ_2 are coefficients calculated through RLS-VDF, and M_h provides the variation range–uncertainty–of the output $\hat{Y}_h$.

Consider an input X_h, a prediction model denoted by $\mathcal{M}$, its output denoted by $\hat{Y}_h = \mathcal{M}(X_h)$, the uncertainty quantification procedure denoted by $\mathcal{U}$ and an interval $[\underline{Y}_h, \overline{Y}_h]$ selection procedure denoted by $\mathcal{S}$, figure 5 depicts the RUL uncertainty quantification uncertainty through HI's error.

Given a particular reference iteration $h = m$ and a HI input $X_h \mid h = m$, consider its first estimated output $\hat{Y}_h = \hat{x}_{h+1} = \mathcal{M}(X_h)$. The subsequent input and its respective output are estimated as follows.

$$\hat{X}_{h+1} = (x_{h-l+1}, \ldots, \hat{x}_{h+1}),\ \hat{Y}_{h+1} = \hat{x}_{h+2} = \mathcal{M}(\hat{X}_{h+1}) \mid h = m \tag{49}$$

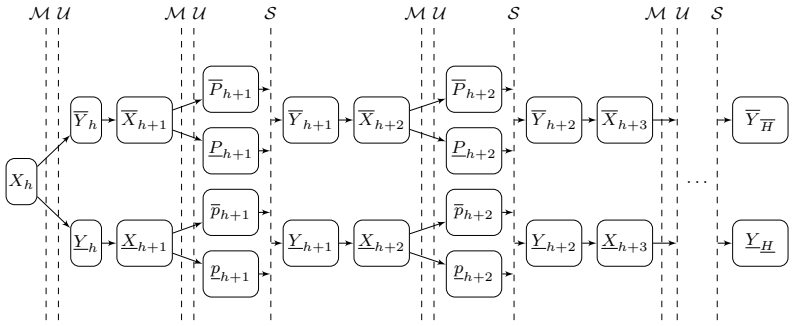


Figure 5: RUL uncertainty through HI's error propagation.

By continuously predicting HI values and using them as following inputs, the following statement is established for any prediction ahead of iteration $h = m$.

$$\hat{X}_{h+t} = (x_{h-l+t}, \ldots, \hat{x}_{h+t}),\ \hat{Y}_{h+t} = \hat{x}_{h+t+1} = \mathcal{M}(\hat{X}_{h+t}) \mid h = m,\ t \geq 0 \quad (50)$$

Therefore, a sequence of predictions

$$(\hat{Y}_{h+t}, \hat{Y}_{h+t+1}, \ldots, \hat{Y}_H) = (\hat{x}_{h+t+1}, \ldots, \hat{x}_{H+1}) \mid h = m,\ H \geq h,\ t \geq 0 \quad (51)$$

until an iteration H when $\hat{Y}_H \leq \text{EoL}$ and $\hat{\text{RUL}} = H - h$. The proposed uncertainty quantification approach transforms this sequence of singular predictions into a series of interval predictions

$$([\hat{Y}]_{h+t}, [\hat{Y}]_{h+t+1}, \ldots, [\hat{Y}]_H) \mid h = m,\ H \geq h,\ t \geq 0 \quad (52)$$

and also elevates the RUL prediction to an interval $[\overline{\text{RUL}}, \underline{\text{RUL}}]$.

During an iteration $(h = m + t) \mid t = 0$, an input X_h provides the interval output $[\underline{Y}_{h+t}, \overline{Y}_{h+t}] = [\underline{x}_{h+t+1}, \overline{x}_{h+t+1}] \mid t = 0$, which can be used to estimate the following interval input $[\underline{X}_{h+t}, \overline{X}_{h+t}] \mid t = 1$ and respective output $[\underline{Y}_{h+t}, \overline{Y}_{h+t}] \mid t = 1$, analogous to (50) input as follows.

$$\underline{X}_{h+t} = (x_{h-l+t}, \ldots, \underline{x}_{h+t+1}),\ \underline{Y}_{h+t} = \underline{x}_{h+t+1} = \mathcal{M}(\underline{X}_{h+t}) \mid t \geq 1 \quad (53)$$

$$\overline{X}_{h+t} = (x_{h-l+t}, \ldots, \overline{x}_{h+t+1}),\ \overline{Y}_{h+t} = \overline{x}_{h+t+1} = \mathcal{M}(\overline{X}_{h+t}) \mid t \geq 1 \quad (54)$$

From the second iteration $(h = m + t) \mid t \geq 1$, because all inputs become intervals $[\underline{X}_{h+t}, \overline{X}_{h+t}] \mid t \geq 1$, consider $[\underline{p}_{h+t}, \overline{p}_{h+t}]$ the interval output of

the lower input $\underline{X}_{h+t}$, $[\underline{P}_{h+t}, \overline{P}_{h+t}]$ the interval output of the upper input $\overline{X}_{h+t}$ and $P_h = \{\underline{p}_{h+t}, \overline{p}_{h+t}, \underline{P}_{h+t}, \overline{P}_{h+t}\}$ the set of predictions at the current iteration. The definitive interval output $[\hat{Y}_{h+t}] = [\underline{Y}_{h+t}, \overline{Y}_{h+t}]$ is established as the global minimum and global maximum of P_h.

$$\underline{Y}_{h+t} = \begin{cases} \min\left(\hat{Y}_h - \mathrm{M}_h, \hat{Y}_h + \mathrm{M}_h\right), \text{ if } t = 0, \\ \min\left(\underline{p}_{h+t}, \overline{p}_{h+t}, \underline{P}_{h+t}, \overline{P}_{h+t}\right), \text{ otherwise.} \end{cases} \tag{55}$$

$$\overline{Y}_{h+t} = \begin{cases} \max\left(\hat{Y}_h - \mathrm{M}_h, \hat{Y}_h + \mathrm{M}_h\right), \text{ if } t = 0, \\ \max\left(\underline{p}_{h+t}, \overline{p}_{h+t}, \underline{P}_{h+t}, \overline{P}_{h+t}\right), \text{ otherwise.} \end{cases} \tag{56}$$

The described procedure is repeated until $\underline{Y}_{h+t}$ and $\overline{Y}_{h+t}$ reaches EoL at instants $\underline{H} = h + t \mid \underline{Y}_{h+t} \leq \text{EoL}$ and $\overline{H} = h + t \mid \overline{Y}_{h+t} \leq \text{EoL}$, respectively. The minimum and maximum remaining useful life are calculated as follows.

$$\underline{\text{RUL}} = \underline{H} - h, \quad \overline{\text{RUL}} = \overline{H} - h \tag{57}$$

4.4. Proposed algorithm

The algorithm 1 depicts the pseudo-code of EGRNN for RUL prediction usage. The models assimilate inputs DSI Z_h, HI X_h, and the output Y_h as well. During the first iteration $h = 1$, the model creates the first IG and do not provide any prediction.

From the second iteration, each IG evaluates the input Z_h by checking if its eccentricity holds (**??**), triggering information assimilation or a new granule creation. After the granulation procedure, the model is used to calculate the output prediction $\hat{Y}_h$ and its interval $\left[\underline{Y}_h, \overline{Y}_h\right]$ multiple steps ahead until each predicted value reach the EoL, finally providing the RUL in an interval $\left[\underline{\text{RUL}}, \overline{\text{RUL}}\right]$.

5. Application to Prognostics of Rolling Bearing

This section aims to evaluate the model's performance in bearing fault prognostics through a series of simulations using accelerated lifetime data from bearing datasets. The primary objectives are to demonstrate the interpretability of the model by analysing the generated IGs, assess the effectiveness of RNN as a predictor within the EM framework, and examine the

Algorithm 1 EGRNN RUL Predictor

Initialize: $c = h = 1, \hat{\text{RUL}} = \underline{\text{RUL}} = \overline{\text{RUL}} = 0, m, \rho, \text{EoL}$
Input: Z_h, X_h, Y_h
Create granule $\mathcal{G}_c$ and do $c = c + 1$
if $h \geq 2$ **then**
  $Create = 1$
  for all granules $\mathcal{G}_i$ **do**
    Calculate $\mathcal{P}_i$ using (**??**)-(**??**)
    Calculate $\hat{Y}_{ih}$ using (36)
    if equation (**??**) holds **then**
      $Create = 0$
    end if
  end for
  if $Create = 1$ **then**
    Create granule $\mathcal{G}_c$ and do $c = c + 1$
  end if
  for all pairs of granules $\mathcal{G}_i$ and $\mathcal{G}_j$ **do**
    if equation (**??**) or (**??**) holds **then**
      Merge $\mathcal{G}_i$ and $\mathcal{G}_j$
    end if
  end for
  Do $t = 0$
  Calculate $\hat{Y}_{h+t}$ using (50)
  Calculate $[\underline{Y}_{h+t}, \overline{Y}_{h+t}]$ using (55)-(56)
  for all granules **do** $\mathcal{G}_i$
    Update W_i, V_i, U_i with (**??**)-(34)
  end for
  while $\hat{Y}_{h+t} \geq$ EoL **do**
    $\hat{\text{RUL}} = \hat{\text{RUL}} + 1$ and $t = \hat{\text{RUL}}$
    Calculate $\hat{X}_{h+t}$ and $\hat{Y}_{h+t}$ using (50)
  end while
  while $\underline{Y}_{h+t} \geq$ EoL **do**
    $\underline{\text{RUL}} = \underline{\text{RUL}} + 1$ and $t = \underline{\text{RUL}}$
    Calculate $\underline{X}_{h+t}$ and $\underline{Y}_{h+t}$ using (53)
  end while
  while $\overline{Y}_{h+t} \geq$ EoL **do**
    $\overline{\text{RUL}} = \overline{\text{RUL}} + 1$ and $t = \overline{\text{RUL}}$
    Calculate $\overline{X}_{h+t}$ and $\overline{Y}_{h+t}$ using (54)
  end while
end if
Output: $\underline{\text{RUL}}, \text{RUL}, \overline{\text{RUL}}$

proposed uncertainty quantification approach. Additional objectives include evaluating the continuous learning process—specifically, the modification of parameters and structure .

5.1. Experimental Setup and Procedures

Table 2: Part of database used

Operational conditions	Identity	Number of files	Life-time (hh:mm)	Fault element
35.0Hz 12kN	B1_1	122	02:03	Outer race
35.0Hz 12kN	B1_2	161	02:41	Outer race
35.0Hz 12kN	B1_3	158	02:38	Outer race
37.5Hz 11kN	B2_2	161	02:41	Outer race
37.5Hz 11kN	B2_4	42	00:42	Outer race

This experiment utilizes data from the XJTU-SY Bearing Datasets [20], which contain comprehensive life-cycle records of 15 rolling element bearings. In particular, the major feature of this dataset is the specification of bearing fault existing at EoL. For this work, it its considered the outer race fault bearings listed on table 2 for assessing the proposed approach, also each bearing's identity is abbreviated in the form Bearing1_1 = B1_1.

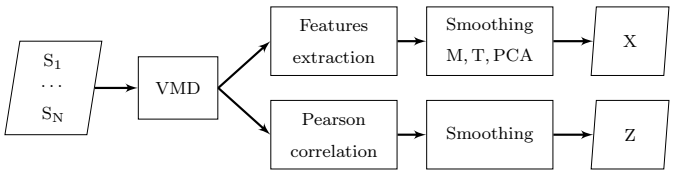


Figure 6: HI and DSI calculation procedure.

The data is submitted to the consecutive transformation procedures depicted by the flowchart in Figure 6. Each sample S is preprocessed by a signal filtering through VMD, followed by the selection of the first intrinsic mode function. Subsequently, the DSI (denoted by $\mathrm{Z} = z_1, z_2, \dots$) is calculated through Pearson's correlation (42) coefficient between the first data sample and the subsequent samples and then smoothed by a sliding EM with a smoothing factor $\alpha = 0.015$. The features are extracted and smoothed by a Exponential Moving Average (EMA) with $\alpha = 0.015$ and then selected based on the monotonicity criterion $\mathrm{M}_i(\mathcal{C}) \geq 0.5$ and trendability $\mathrm{T}_i(\mathcal{C}, t) \geq 0.5$.

Table 3: Features list [21].

Time Domain	Mean, **Absolute Mean**, **Variance**, **Standard Deviation**, Skewness, **Entropy**, **RMS**, **Maximum**, **Peak-to-peak**
Frequency Domain	**Spectral Energy**, **Energy Mean**, Energy Variance, Energy Standard Deviation, Energy Skewness, Energy Entropy, Energy Kurtosis, Energy RMS, **Energy Density**

Table 3 presents the features used in this work, among which those highlighted in bold were selected for this study, which were also employed in [21] for the same purpose. Finally, the features holding monotonicity and trendability criteria are fused by PCA into the HIs (denoted by $\mathrm{X} = (x_1, x_2, \dots, x_N)$). The EGRNN is then simulated with each bearing's HI time series X and its respective associated DSI time series Z.

5.2. Choice Hyperparameters

Although the EGRNN incorporates auto-tuning parameters and structural modifications inherent to an EM, some parameters, as listed in Table 4, require fine-tuning in an offline pre-training step. This is due to their inability to adapt during each iteration of the learning process.

The model's hyperparameters are fine-tuned by minimizing the time-weighted Mean Absolute Percentage Error (MAPE) of the RUL predictions,

Table 4: Offline tuned parameters.

Parameter	Symbol	Description
Window length	l	Quantity of previous steps in X_h
Granulation sensitivity	m	Permeate the quantity of IGs created
Granularity	$\mathcal{E}$	Amount of information within an IG
Granularity's optimizer	G	Product between $\overline{cov}$ and $\overline{sp}$
Hidden length	$\mathrm{n_h}$	Quantity of hidden units if h_h
Learning rate	η	Size of algorithm's steps when adjusting the neural network parameters

Table 5: MAPE results.

Identity	B1_1	B1_2	B1_3	B2_2	B2_4
MAPE	398.78	643.93	114.10	577.40	22.41

where RUL_h and $\hat{\mathrm{RUL}}_h$ are, respectively, the current real and predicted RUL values and h_{EoL} is the instant when RUL reaches the EoL.

$$\mathrm{MAPE}_h = \frac{1}{(h - h_{EoL})} \sum_{k=h}^{h_{\mathrm{EoL}}} \left| \frac{\mathrm{RUL}_k - \hat{\mathrm{RUL}}_k}{\mathrm{RUL}_k} \right| \tag{58}$$

Additionally, the granularity ρ is optimized by maximizing $G(\rho)$. Both the hyperparameters and granularity are optimized using the Optuna framework, which supports hyperparameter optimization through methods such as Bayesian optimization and random search.

5.3. Experimental Results

This section presents and discusses the simulations of EGRNN applied to RUL predictions on XJTU bearing datasets, whose results are evaluated by the proposed MAPE established by (58).

Primarily, the DSI time series Z is created by (42) and the HI time series X is created by fusing the list of selected features on table 3 with PCA. Sequentially, by fixing a window length $l = 5$ and a $x_{\mathrm{EoL}} = 0.3$ the parameters of table 4 area optimized.

Table 5 provides the MAPE results computed from the last 85% of predicted RUL values. Notice that EGRNN was able to ensure MAPE values below 50% only for B2_4, while the others bear more challenging data, resulting in higher MAPE for any parameter combination.

Figures 7 a)-e) upper depicts the granulation over DSI of each bearing, where it is possible to observe that the DSI at the end of the life-cycle is separated in a different plateau from the DSI at the beginning of the life-cycle. Even though all bearings present the same fault at EoL, the DSI series are slightly different from each other. All bearings presents their fault stages with DSI values around 0, but their previous DSs do not share the same range of DSI and it is also noticeable different dispersions patterns.

Even though bearings B1 and bearings B2 work at different rotations and loads (35.0Hz|12kN and 37.5Hz|11kN respectively), there is no evidence of a relationship between DS patterns and the operational condition. While B1_1 and B1_3 seems to gradually change their DSs, B1_2, B2_2 and B2_4 got abruptly into the final DS.

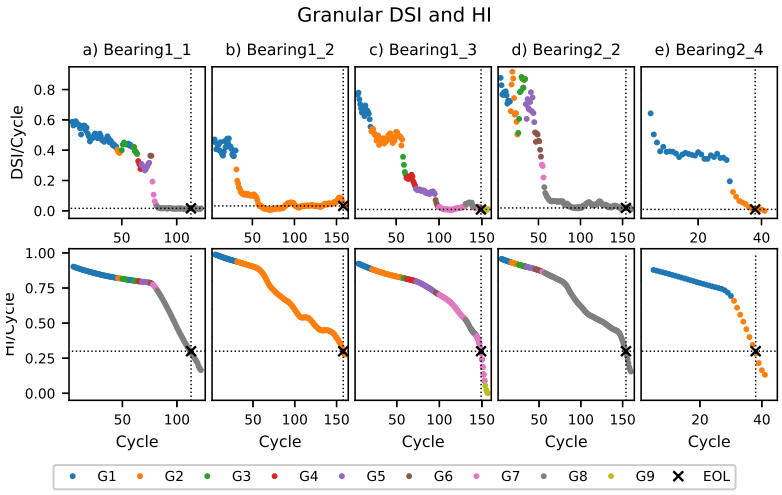


Figure 7: Bearings DSI and HI time series granulated over time.

It is crucial to highlight that each bearing's granules $G1, \ldots, G8$ are completely independent (the IGs G1 from each bearing are not the same). Additionally, it is possible to verify the creation or merge of granules related to different patterns of DS. The optimal sensibility factor m was not able to standardize the amount of IGs created, resulting in 2 granules for B1_2 and B2_4 and more than 4 granules for B1_1, B1_3 and B2_2.

Figures 7 a)-e) lower depicts the bearing's granulated HI, which is a consequence of DSI granulation. As highlighted before, despite the same fault,

the HI degradation behavior is also slightly different for each bearing. However, it is important to mention that all HI's curves have a transition point from where the degradation process becomes faster (its slope increases), even though their shapes vary.

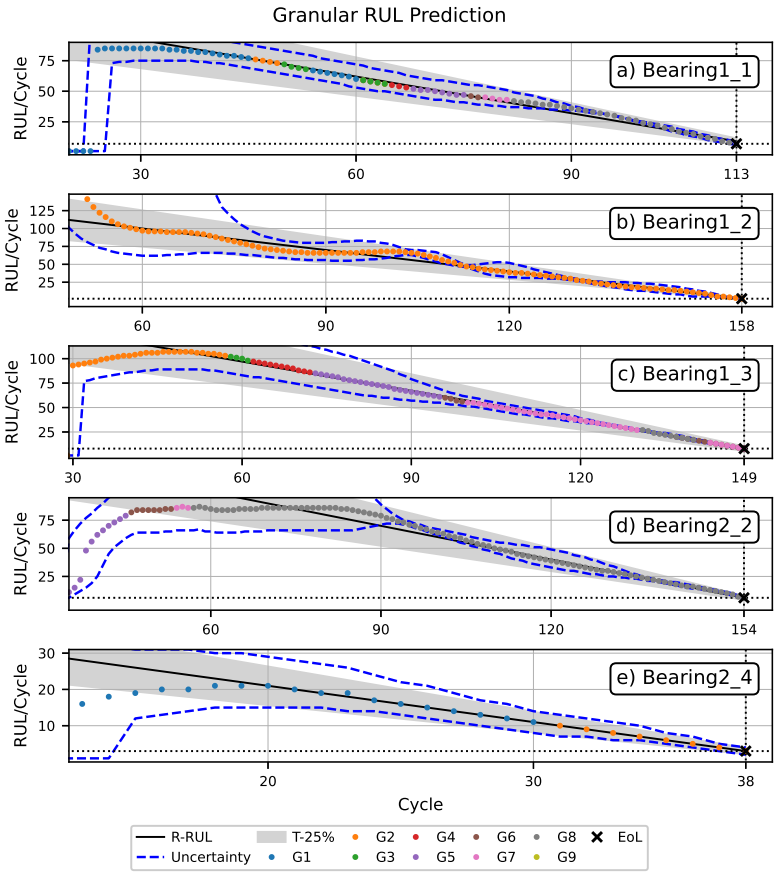


Figure 8: Bearings RUL prediction granulated over time.

Bearings B1_1 and B2_4 present a quite similar HI degradation behavior, but while the first lasts more than 100 cycles until reaching EoL, the second lasts only 40. Bearings B1_2, B1_3 and B2_2 lasts about 150 cycles but present HI significantly different curves. Therefore, it is possible to associate them and conclude that the degradation patterns are connected to the respective DS. The last granule created or activated triggers the HI during its faster degradation. Figures 8 a)-e) depict the granulated RUL prediction,

also a consequence of DSI granulation. The colored dots denotes the RUL predictions and their IGs, gray shade represents the $\alpha - \lambda$ accuracy cones of $\pm 25\%$, and the region between dashed blue lines represents the uncertainty zone of $\hat{\text{RUL}}_h$.

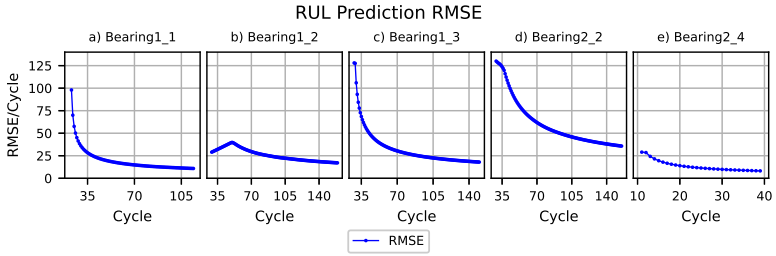


Figure 9: Bearings RUL prediction RMSE.

By comparing the real RUL predicted $\hat{\text{RUL}}$ and analyzing their Root Mean Square Error (RMSE) over time (Figure 9), the continuous learning feature of EGRNN becomes evident. The RLRNN approach handled the nonlinearity of the time series HI, and the predicted values of RUL were closer to the real values during the consecutive iterations. The uncertainty was quantified based on the granularity ρ and allowed the RUL prediction to be made under a confidence interval, which becomes narrow near the EoL.

6. Conclusions

This paper presents the EGRNN, a novel evolving model for time series processing, which combines the TEDA framework for data-stream analysis with the approximation power of the RNN for modeling nonlinear behaviors in non-stationary time series. The EGRNN is employed for the particularly challenging task of fault prognostics in rolling bearings, combining the PJG uncertainty modelling for obtaining confidence intervals of the RUL predictions. The results indicate the potential of applying evolving granular models for fault prognostics, considering the inherent complex behavior of the degradation phenomena.

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